

Shan Shaji

Idukki, Kerala

+91-9778075821 shanshajisharon@gmail.com
linkedin.com/in/shan-shaji github.com/Shanios

PROFILE SUMMARY

Entry-level Software Developer with hands-on experience building full-stack web applications and Python-based backend systems. Proven ability to design and deploy end-to-end solutions using FastAPI, React, Node.js, and cloud platforms. Comfortable with REST API design, database integration, real-time communication, and third-party API integration. Eager to contribute and grow long-term within a software engineering team.

EDUCATION

- Bachelor of Technology (B.Tech) in Computer Science** 2026
APJ Abdul Kalam Technological University, Kerala CGPA: 7.43

SKILLS SET

- Languages:** Python, JavaScript, SQL
- Backend:** FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy
- Frontend:** React.js, HTML5, CSS3, Tailwind CSS
- Databases:** PostgreSQL, MySQL, MongoDB
- Tools & Platforms:** Git, GitHub, VS Code, Railway, Vercel, Supabase
- Libraries & APIs:** PyTorch, OpenCV, MediaPipe, scikit-learn, WebRTC, Twilio API, Gemini API, JWT
- Soft Skills:** Problem-solving, Logical thinking, Adaptability, Communication

EXPERIENCE

- Software Engineering Intern** Mar 2024 – Apr 2024
Luminar Technolab, Kochi, Kerala

- Developed Python backend utilities to process and validate student performance data, reducing processing errors by 20%.
- Debugged Python scripts and SQL queries, identifying and resolving logical and runtime errors during testing phases.
- Refactored scripts into modular components, improving readability and maintainability.

PROJECTS

- Steins Gate — AI Smart Surveillance Platform** *PyTorch, YOLOv8, FastAPI, PostgreSQL, React, WebSockets*

- Built a real-time anomaly detection pipeline using ResNet18 + Transformer for CCTV stream analysis, achieving 98.67% classification accuracy across intrusion and suspicious activity events.
- Implemented fall detection using YOLO-Pose and LSTM with post-fall immobility analysis for elderly safety monitoring, reducing false positives through temporal sequence modeling.
- Developed contactless heart rate estimation using Remote Photoplethysmography (rPPG) and Eulerian Video Magnification (EVM) from live facial video feeds with real-time BPM tracking.
- Integrated WebSocket-powered alert system with Twilio WhatsApp messaging and automated voice call escalation for real-time emergency notifications to caregivers and security personnel.

- Student Engagement Platform** *FastAPI, React, PostgreSQL, Supabase, OpenCV, MediaPipe, scikit-learn, Groq*

- Built a full-stack platform for real-time student engagement monitoring in online classes using a webcam-based ML pipeline (OpenCV + MediaPipe + scikit-learn) for facial landmark analysis and engagement scoring.
- Developed a RAG chatbot using Groq LLM with embeddings stored on Supabase Storage, enabling context-aware Q&A for course material.
- Designed RESTful API endpoints with FastAPI and SQLAlchemy for session management, attendance tracking, and engagement analytics; deployed backend on Railway and frontend on Vercel.
- Built session analytics and reporting module generating post-session engagement summaries with WhatsApp export format via Twilio API.

ADDITIONAL INFORMATION

- Strong interest in backend and full-stack software development.